SimplyMZ

An MCP server that lets an AI assistant analyse LC-MS / GC-MS runs with deterministic tools and an evidence ledger

How to use it: summary, use cases, sample files, specifications

Version 0.1.0 pre-release · GPLv3 · Research use only

START HERE

Summary

SimplyMZ connects an AI assistant (Claude Desktop, Claude Code, or any MCP client) to your mzML files. You ask in plain language; the assistant calls SimplyMZ's tools; the tools do the chemistry and record every result.

What it answers

- Is this run healthy?
- Did the batch pass QC?
- What differs between two groups?
- What is this peak?

Why you can trust it

- AI plans and explains; tools compute
- Every result in an append-only ledger
- Reports only contain traceable numbers
- Says "insufficient evidence" when it is

What's inside

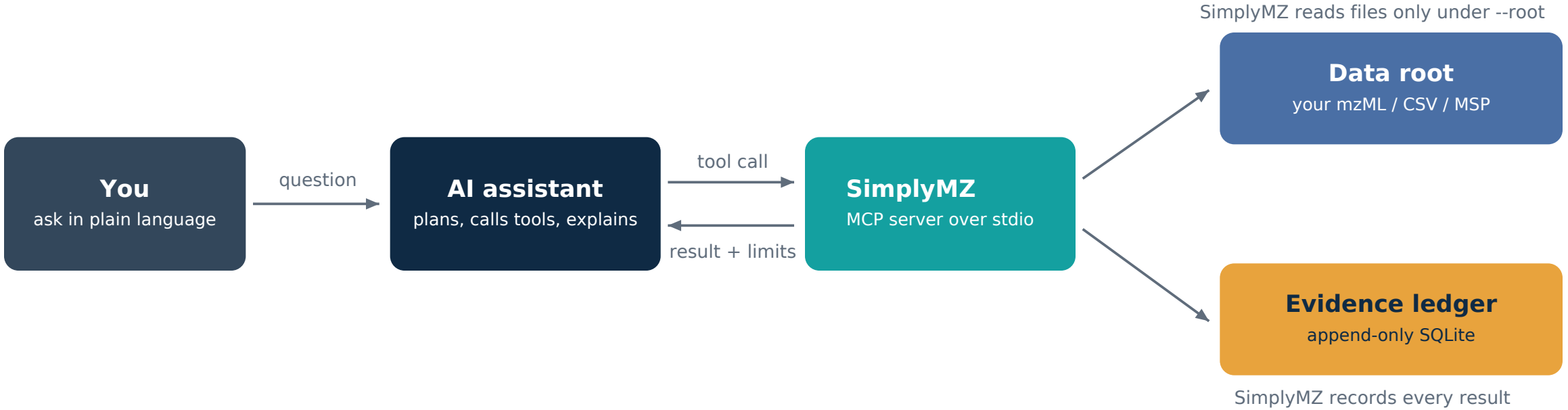
- 19 tools, 4 guided prompts, 2 resources
- 10 QC metrics with pass/warn/fail
- Formula, isotope and library matching
- Markdown/HTML reports with provenance

Where it stands

- 458 automated tests; 97.6% coverage
- All 6 milestone gates pass locally
- Validated mainly on synthetic data
- Pending: CI, PyPI release, LLM eval

You need: uv, mzML files (convert vendor files with ProteoWizard msconvert), and an MCP client. Sample data ships in `tests/fixtures/`.

How it works



- **Division of labour:** the assistant decides what to do and explains; SimplyMZ's deterministic tools compute; the ledger remembers every result with its parameters and input file hashes.
- **Every tool returns the same envelope:** status (ok / insufficient_evidence / error), a one-sentence verdict, data, limits, next steps, and a provenance id (ev:…).
- **Handles, not raw data:** tools pass short ids (run:, batch:, chrom:, ftab:, lib:, job:) instead of peak arrays, so answers stay small (a token budget per tool).

Install and connect

- Install **uv** (docs.astral.sh/uv). Python 3.11+ is fetched automatically.
- Convert vendor files (.raw, .d, .wiff, .wiff2, .lcd) to mzML with ProteoWizard `msconvert`; SimplyMZ refuses vendor formats with this hint.
- Pick a **data root**: the only folder SimplyMZ may read. State and cache go to `<root>/.simplymz`; reports to `<root>/simplymz_reports/`.

Claude Desktop (claude_desktop_config.json)

```
{
  "mcpServers": {
    "simplymz-mcp": {
      "command": "uvx",
      "args": ["simplymz-mcp", "--root", "/path/to/mzml"]
    }
  }
}
```

Claude Code

```
claude mcp add --transport stdio simplymz-mcp \
  -- uvx simplymz-mcp --root /path/to/mzml
```

Before the PyPI release (from a checkout)

```
uvx --from /path/to/simplymz-mcp simplymz-mcp \
  --root /path/to/mzml
```

Optional: `uvx --with "matchms>=0.33,<0.34" simplymz-mcp …` enables `engine="matchms"` in match_spectrum as an independent cross-check (same scores as the built-in engine).

Check it works: ask the assistant to call `server_info`; it reports the version, data root, limits and tool list.

Sample files: where to find them

Copy `tests/fixtures/` from the repository to a folder and use it as `--root`. All files are synthetic (CC0) with known ground truth in matching `*.truth.json` files.

Location	Contents	Use it for
`runs/clean_pos.mzML`	12 metabolites, positive mode, MS1, 201 scans (plain mzML)	Explore a run, EIC, integration
`runs/clean_neg.mzML.gz`	5 compounds, negative mode (gzipped)	Negative-mode adducts
`runs/ms2_dda.mzML.gz`	8 compounds with DDA MS2 spectra; kynurenine is not in the library	Identify a peak
`batch_qc/ + sample_sheet.csv`	11 injections: 2 blanks, 3 pooled QCs, 3 × group A, 3 × group B; 5 compounds differ between A and B	Batch QC, compare groups
`faults/*.mzML.gz`	One injected fault each: rt_shift, mass_error, intensity_scale, contaminant, dropout, carryover_blank	See QC and run cards react
`library/synthetic_library.msp`	11 synthetic MS/MS spectra, CC0 licence	match_spectrum
`fetch_public_dataset("openms-ffm-1")`	Real LC-MS file from OpenMS (BSD-3, 1.1 MB, checksum-pinned); asks before downloading	Real-data test drive
`scripts/download_library.py`	Opt-in MassBank download with per-record licences (~670 MB)	Real spectral library

PART 2

Use cases

Six things you can ask, step by step, with the sample files

Use case 1: Explore a run

1. `load_run` — Path under the root; same file → same handle
2. `run_card` — Metadata, 40-point TIC, MS2 count, anomalies
3. `get_chromatogram` — TIC, BPC or EIC (m/z ± ppm) with peaks
4. `integrate_peak` — Area, height, FWHM, SNR + precision note

Ask (prompt /explain_chromatogram, or plain words):

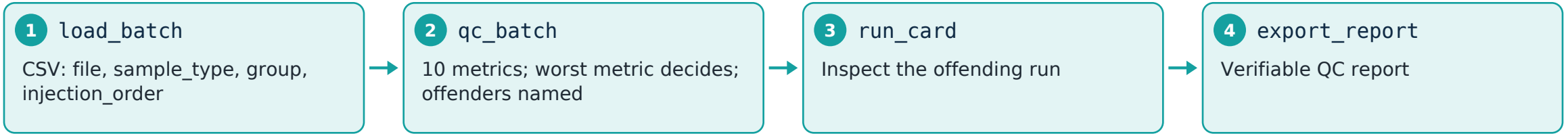
```
Load runs/clean_pos.mzML, show its run card, then the EIC of caffeine
[M+H]+ at m/z 195.0877 and integrate the main peak.
```

- **Anomalies flagged:** RT gaps, empty scans, TIC dropouts, polarity switching (try `faults/dropout.mzML.gz`: 12 empty scans at 4.0 min).
- **Every area comes with a precision note:** within ±3% at SNR ≥ 100; about 1.5% SD at SNR 50–100; warnings for overlapping peaks, few points across a peak, or peaks cut off at the end of a run.
- **No signal?** You get *insufficient_evidence* with reasons (absent, below detection, wrong polarity, outside the ppm window), not a guess.

Real result on the sample files (tool verdicts, verbatim)

```
run_card: runs/clean_pos.mzML: 201 spectra, positive mode, no acquisition anomalies detected.
get_chromatogram: EIC m/z 195.0877 +/- 10 ppm: 1 peak detected, largest at 4.4 min.
integrate_peak: Peak at 4.4 min: area 476004 counts*min, height 3.02755e+06 counts, undefined SNR; area precision not estimable (no
measurable noise).
```

Use case 2: Did the batch pass QC?



Ask (prompt /qc_batch):

```
/qc_batch sample_sheet=batch_qc/sample_sheet.csv
```

- **Reference features** are picked automatically from the QC injections (stable, isolated, high-intensity), or pass `targets=[{mz, rt_min}]` for **absolute** mass accuracy.
- **Thresholds** are configurable defaults (next slide), not standards; override per call with `thresholds={"rt_drift_min": {"warn": 0.05, "fail": 0.1}}`.
- **Validated by fault injection:** each fault at 2× its threshold fails its metric (100%); at 0.5× nothing fails (0%). No QC injections → *insufficient_evidence*.
- Up to 60 runs per call; split bigger sample sheets.

Real result on the sample files (tool verdicts, verbatim)

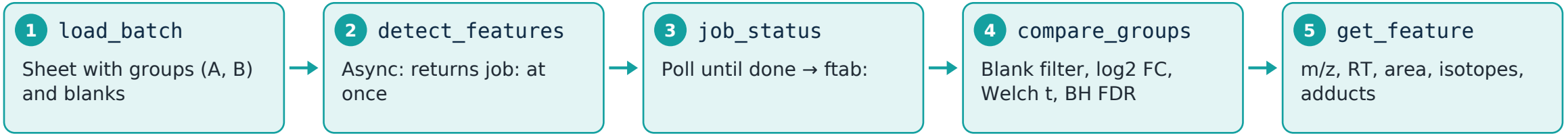
```
qc_batch (batch_qc/sample_sheet.csv): Batch PASSES QC.
Swap in a run from faults/ (e.g. mass_error, dropout) to see the matching metric warn or fail with the offending run named.
```

The 10 QC metrics

Metric	What it measures	Warn above	Fail above
`tic_cv_qc_pct`	CV of total MS1 TIC across QC injections	15 %	25 %
`rt_drift_min`	Largest RT deviation of a reference feature from its QC median	0.1 min	0.2 min
`mass_error_ppm`	Largest per-run median m/z error of reference features	5 ppm	10 ppm
`qc_feature_cv_pct`	Median CV of reference feature areas across QCs	20 %	30 %
`missingness_pct`	Reference features not detected (non-blank runs)	5 %	20 %
`blank_carryover_pct`	Largest blank / preceding-injection area ratio	1 %	5 %
`blank_contamination_pct`	Largest blank TIC as % of median sample TIC	15 %	30 %
`ms2_count_cv_pct`	CV of MS2 spectrum counts across QCs	10 %	20 %
`intensity_trend_pct_per_injection`	Largest QC area trend per injection	2 %	5 %
`scan_dropouts`	Longest run of empty or TIC-dropout MS1 scans	0 scans	4 scans

Verdict rule: value ≤ warn → **pass**; warn < value ≤ fail → **warn**; value > fail → **fail**. Metrics that cannot be computed (e.g. no MS2) are *not_applicable*. Defaults live in `qc_defaults.toml`.

Use case 3: What differs between groups?



Ask (prompt /compare_groups):

```
/compare_groups sample_sheet=batch_qc/sample_sheet.csv group_a=A group_b=B
```

- **Defaults:** blank_fold 3 (features not 3× above blanks are removed), min_presence 0.5, top_n 10; feature m/z tolerance 10 ppm, alignment RT tolerance 0.15 min.
- **On the sample batch:** all 5 planted differences (adenosine ×4, tyrosine ×0.25, caffeine ×5, kynurenine ×3, hippuric acid ×4) rank in the top 10; the blank contaminant is filtered out.
- **Honest statistics:** with n = 3 per group the result carries a small-sample caveat; no biological claims are made.
- Long jobs can be stopped with `cancel_job`; results survive a restart.

Real result on the sample files (tool verdicts, verbatim)

```
compare_groups: B vs A (n=3 vs 3): 5 of 12 features differ at q < 0.05; top C00004 m/z 268.1040 at 2.90 min, log2 FC +1.95.
(adenosine: planted 4×, log2 = 2)
```

Use case 4: What is this peak?

1. `propose_formulas` — Formulas within 5 ppm, ranked by isotope fit
2. `load_library` — MSP/MGF library → lib: handle
3. `match_spectrum` — Cosine / modified cosine; licence shown
4. `assign_confidence` — Level from ledger evidence only

Ask (prompt /identify_peak):

```
/identify_peak path=runs/ms2_dda.mzML.gz mz=195.0877 rt_min=4.4
               library=library/synthetic_library.msp
```

Level	Meaning (Schymanski et al. 2014)	How SimplyMZ assigns it
1	Confirmed structure	Never in v0.1 (needs a reference standard)
2	Probable structure	Library match passes (score ≥ 0.7, ≥ 2 fragments) and agrees with formula
3	Tentative candidate	Near-miss library hit (≥ 0.5) or library/formula conflict
4	Unequivocal formula	Exactly one formula fits mass and isotopes
5	Exact mass	Only an accurate m/z

Real result on the sample files (tool verdicts, verbatim)

```
caffeine 195.0877 @ 4.4 min: best C8H10N4O2 (unequivocal) · library score 0.9956, 2 fragments, CC0 · Level 2: caffeine
kynurenine 209.0921 @ 5.6 min: 2 formulas consistent · no library spectrum within tolerance · Level 5 (exact mass)
```

Use case 5: Verifiable reports

Ask for a report at the end of any workflow. The assistant writes the narrative; numbers come from **slots** filled from the ledger:

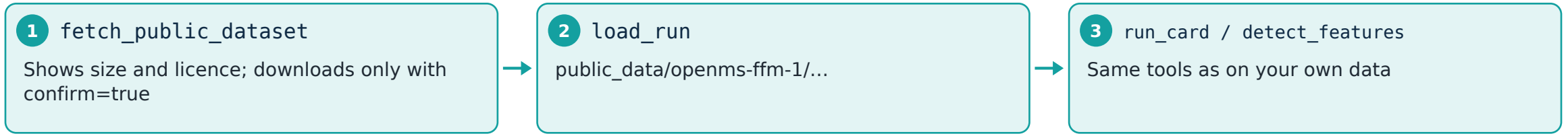
```
export_report(title="Batch QC", filename="qc_oct06",
  sections=[{"heading": "QC", "body":
    "Verdict {ev:01J...:data.overall}; RT drift
     {ev:01J...:data.metrics.rt_drift_min.value} min."}])
```

Number guard

- Any number in the text that is not in referenced evidence is rejected, and listed
- "Recovery was 98%" → rejected
- Formulas, adducts, 13C, run names are not treated as measurements

- **Output:** `simplymz_reports/<name>.md` and `.html`; existing files are not overwritten unless you ask.
- **Limits and disclosures** from every cited result are copied in automatically (e.g. the peak-area precision note, the matching-engine note).
- **Provenance table:** tool, time (UTC), result digest, input file SHA-256 and software versions for each piece of evidence.

Use case 6: Real public data and libraries



- **Registry only:** exact URL, size and SHA-256 are pinned; HTTPS to allowlisted hosts; files are written atomically and verified. This is the **only** tool that uses the network.
- **openms-ffm-1:** real LC-MS metabolomics excerpt (279 MS1 spectra, positive, RT 33.4–41.6 min), BSD-3-Clause, from the OpenMS test suite.
- **MassBank (opt-in):** `uv run python scripts/download_library.py --root DATA` builds an MSP with each spectrum's own licence (CC BY, CC BY-SA, …) taken from MassBank.json.
- No third-party spectra are bundled; NIST libraries are not supported for redistribution.

Real result on the sample files (tool verdicts, verbatim)

```
fetch_public_dataset("openms-ffm-1") without confirm: Not downloaded: openms-ffm-1 is 1.14 MB (BSD-3-Clause (OpenMS test data)). Ask the user, then call again with confirm=true.
```

PART 3

Specifications and capabilities

Tools, prompts, resources, formats and limits

Tool reference (1/3): Run and chromatogram

Tool	What it does	Token budget
`server_info`	Version, data root, limits, optional extras, tool list	600
`load_run`	Load an mzML/mzML.gz under the root; returns run:	700
`run_card`	Metadata, downsampled TIC, MS2 counts, anomalies	1500
`get_chromatogram`	TIC / BPC / EIC (mz ± mz_tol_ppm, RT range) + peaks; chrom:	1500
`integrate_peak`	Area, height, FWHM, asymmetry, SNR, precision note	900
`release_handle`	Release a handle you no longer need	300

Tool reference (2/3): Batch, features and jobs

Tool	What it does	Token budget
`load_batch`	Load a CSV sample sheet; validates every row; batch:	1500
`qc_batch`	10 QC metrics, verdicts, offenders; targets/threshold overrides	2000
`detect_features`	Async feature detection on a run or batch; job:	400
`job_status`	Progress and result of a job (ftab: when done)	2600
`cancel_job`	Request cancellation of a running job	300
`get_feature`	Feature card: m/z, RT, area, shape, isotopes, adducts	800
`compare_groups`	Blank filter, log2 FC, Welch t, BH FDR, top-N ranking	2000

Tool reference (3/3): Annotation, reports and data

Tool	What it does	Token budget
`propose_formulas`	Formulas for an ion m/z, Seven Golden Rules subset, isotope fit	1500
`load_library`	MSP or MGF library with per-spectrum licences; lib:	600
`match_spectrum`	Cosine / modified cosine vs library; engine builtin or matchms	1500
`assign_confidence`	Schymanski level 2-5 from evidence ids; cannot be set by hand	900
`export_report`	Markdown + HTML with ledger slots, number guard, provenance	800
`fetch_public_dataset`	Pinned public dataset; needs confirm=true	600

Budgets cap each answer's size; oversized lists are truncated with `truncated: true`. Full schemas: `docs/tools.md`.

Guided prompts and resources

Prompt	Arguments	What it does
`/qc_batch`	`sample_sheet`	Batch QC with a metrics table, meaning of findings, next checks
`/compare_groups`	`sample_sheet, group_a, group_b`	Feature detection and ranked group differences with caveats
`/identify_peak`	`path, mz, rt_min, adduct?, library?`	Formulas, library match, confidence level; never raises the level
`/explain_chromatogram`	`path, mz?`	Plain-language run and chromatogram summary

Resource	Content
`simplymz://run/{run_id}/card`	Run card for run:<run_id>
`simplymz://batch/{batch_id}/qc`	QC report for batch:<batch_id> (default thresholds)

Every prompt carries honesty rules: cite tool numbers and provenance ids, repeat the limits, don't speculate beyond listed causes, treat *insufficient_evidence* as a valid answer.

Specifications

Area	Specification
Input formats	mzML, mzML.gz; CSV sample sheets; MSP / MGF libraries
Spectra	Centroided expected (profile flagged); MS1 and MSn; positive, negative, switching
Limits	Files ≤ 2048 MB, ≤ 500,000 spectra (configurable)
Adducts (13)	[M+H]+ [M+Na]+ [M+K]+ [M+NH4]+ [M+H-H2O]+ [M+2H]2+ [2M+H]+ [M-H]- [M+Cl]- [M+HCOO]- [M+CH3COO]- [M-H-H2O]- [M-2H]2-
Formula elements	C H N O P S, optional Cl Br
Default tolerances	EIC 10 ppm · formulas 5 ppm · fragments 0.01 Da · precursor 10 ppm

Area	Specification
Transport	MCP over stdio (MCP Python SDK 2.x)
Handles	run: batch: chrom: feat: ftab: lib: job: ev: (restart-safe)
Storage	SQLite (handles, jobs, ledger) + Parquet run cache keyed by SHA-256
Safety	Reads only under --root; symlink escape blocked; vendor files refused; unknown tool arguments rejected
Network	None, except fetch_public_dataset (registry, confirm=true)
Install	Python ≥ 3.11 · ~101 MB core · GPLv3

Built-in honesty

Evidence ledger

- One immutable row per result
- Parameters, input hashes, versions
- Database blocks edits and deletes

Insufficient evidence

- A valid answer, not an error
- Used when a peak, QC or match is unsupported
- Level 1 identity is never claimed

Disclosures travel

- Precision notes on every area
- Matching-engine note on every match
- Copied into reports automatically

- **Instructive errors:** every refusal says what to do next (e.g. msconvert command, valid adducts for the run's polarity, how to split a large batch).
- **Verified process:** each milestone passed a 9-step gate (lint, types, tests, coverage, contract, acceptance, regression, records, secrets scan).

Validation results

Evaluation	Result	Data
QC: fault at 2× / 4× its threshold detected	100 % (6 fault types)	synthetic
QC: false fail at 0.5× threshold	0 %	synthetic
Formula: true formula among candidates (5 ppm)	100 % of 35 compounds	oracle masses
Formula: ranked first with isotopes (< 400 Da)	97–100 %	simulated isotopes
Library: out-of-library spectra given no identity	100 %	synthetic
Library: decoy false positives	0 %	synthetic
Feature detection recall	100 % synthetic; 5 of OpenMS's 10 most intense on a real file	synthetic + OpenMS
Peak area accuracy	within 3 % at SNR ≥ 100; ~1.5 % SD at SNR 50	synthetic
Spectral similarity vs matchms	identical (max diff 2×10^{-16}, 1,800 pairs)	random spectra

Not yet measured: Claude with vs without SimplyMZ (harness ready in `eval/llm_eval.py`, needs an API key).

Limitations

- **Mostly synthetic validation.** One real public file is used for integration and a recall benchmark; real-data feature detection is weaker (5/10 of OpenMS's top features).
- **Simple feature detector** (masstrace-lite). pyOpenMS is planned as an optional engine.
- **Identification stops at Level 2.** No reference standards, RT prediction or in-silico fragmentation. Formulas use C, H, N, O, P, S, Cl, Br only.
- **QC thresholds are defaults, not standards.** Tune them to your instrument and method.
- **mzML only.** Convert vendor files first; profile data are flagged, not centroided.
- **Research use only**, not for clinical or regulatory decisions.

Troubleshooting

You see	Meaning	Do this
`VENDOR_FORMAT`	A .raw/.d/.wiff/.lcd file	Convert with msconvert, load the .mzML
`PATH_OUTSIDE_ROOT`	Path leaves the data root	Copy the file under --root or restart with another root
`MALFORMED_FILE`	Truncated or not mzML	Re-export; check the copy finished
`POLARITY_MISMATCH`	Adduct sign ≠ run polarity	Use an adduct from the hint list
`UNKNOWN_HANDLE`	Released, expired or file changed	load_run / load_batch again
`BUDGET_EXCEEDED`	Answer too large, or > 60 runs	Narrow the request or split the batch
`insufficient_evidence`	Data do not support an answer	Read limits and next_steps; this is a valid result
`Unknown argument error`	Misspelled or unsupported parameter	Check docs/tools.md

Where to find more

File / folder	What's there
`README.md`	Install, configuration, workflows, benchmarks, limitations
`docs/tools.md`	Every tool's schema, example input and token budget (generated)
`docs/data.md`	Sample data, public dataset, MassBank, licences
`docs/adr/`	Design decisions (feature engine, storage, budgets, libraries, ledger, …)
`docs/progress/`	Milestone logs, gate results, final verification, sign-off
`eval/results.md`	Evaluation tables
`tests/fixtures/`	Sample files with ground truth
`docs/media/DEMO_SCRIPT.md`	Six-scene demo you can follow yourself

Status: feature-complete for v0.1; awaiting CI on GitHub, TestPyPI release, the LLM comparison and human sign-off before the PyPI release.